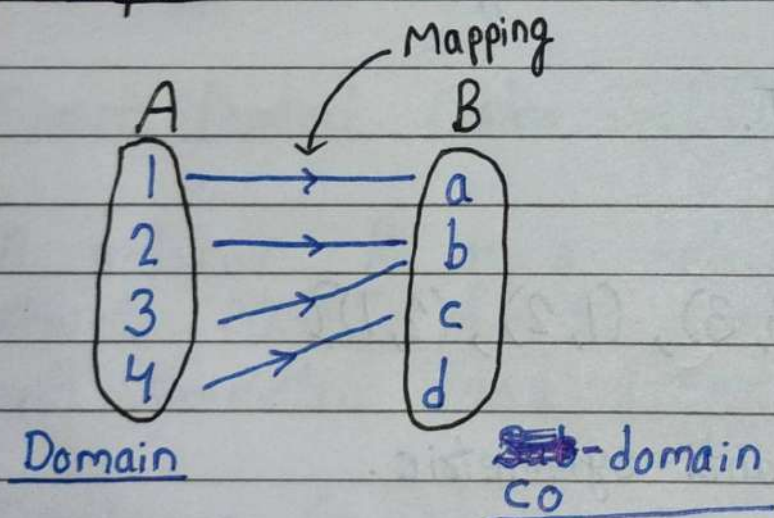


Functions

Functions are widely used in science and in most field of mathematics and research area. It has been said that functions are central object of investigation.

Let, A & B be two non-empty sets. A function f from A to B is a subset of ordered pairs with the property that for each element x in A , there is a unique element y in B such that $(x, y) \in f$.
 $f: A \rightarrow B$.

Example:



Domain:

All the elements which come under set A is known as domain and it is denoted by $\text{dom}(f)$. Its members are the first coordinate of the ordered pairs belonging to f .

For example; $A = \{1, 2, 3, 4\}$.

Co-domain:

All the elements which come under set B is known as co-domain and it is denoted by $\text{codomain}(f)$.

For example; $B = \{a, b, c, d\}$.

Range:

The participated elements of set B in mapping are known as range.

For example; $\text{range} = \{a, b, c\}$.

Q. If $|A| = m$ & $|B| = n$, then how many functions are possible?

Ans. The possible functions are n^m .

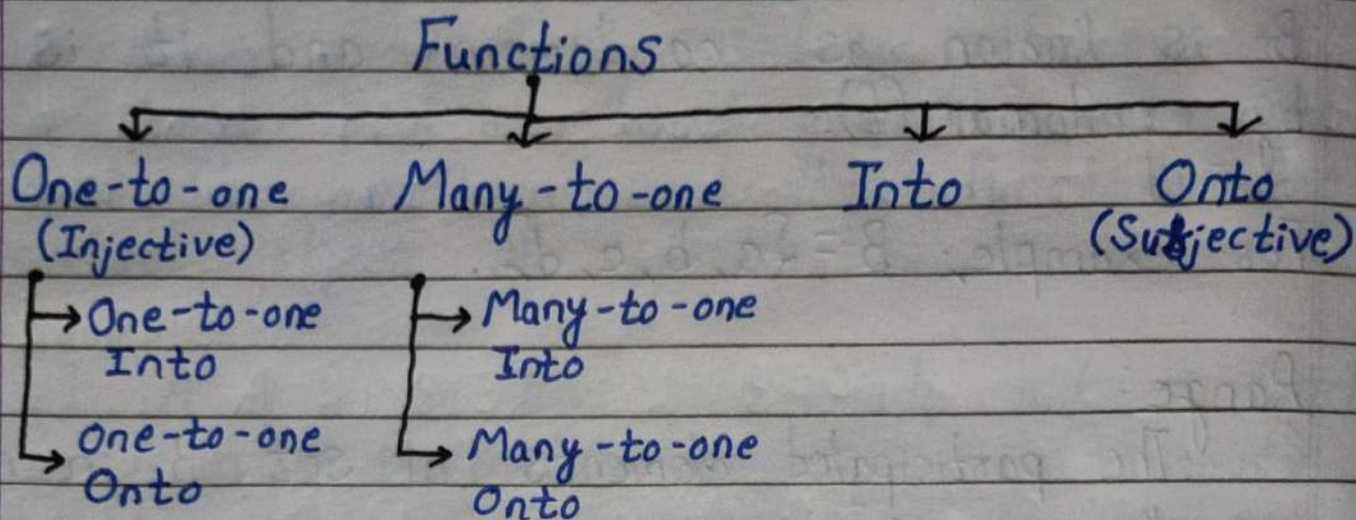
Q. Let, $|X|$ & $|Y|$ is there cardinality. It is given that exactly 97 functions from X to Y , then which of the following is true?

1. $|X| = 1$ & $|Y| = 97$ ✓ $\Rightarrow 97^1 = 97$.

2. $|X| = 97$ & $|Y| = 1$ ✗

3. $|X| = 97$ & $|Y| = 97$ ✗

Types of Functions

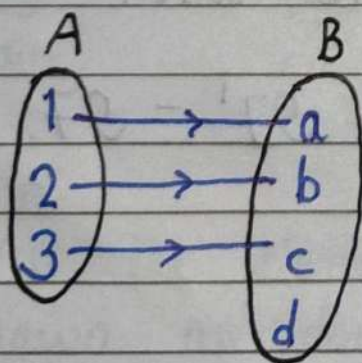


One-to-one (Injective) Function

A function $f: A \rightarrow B$ is called one-to-one function if distinct element of set A are mapped with distinct element of set B .

Example: $A = \{1, 2, 3\}$ & $B = \{a, b, c, d\}$

$$f = \{(1, a), (2, b), (3, c)\}$$

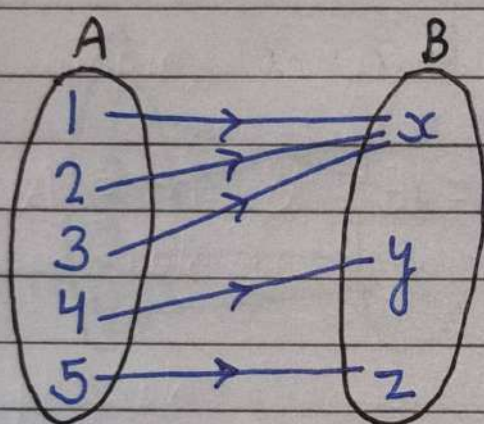


Many-to-one Function

A function $f: A \rightarrow B$ is called many-to-one function if there exist two or more than two different element in A having same image in B .

Example: $A = \{1, 2, 3, 4, 5\}$ & $B = \{x, y, z\}$

$$f = \{(1, x), (2, x), (3, x), (4, y), (5, z)\}$$



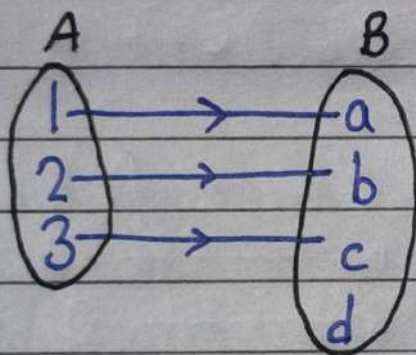
Into Function

A function $f: A \rightarrow B$ is called into function if there exist at least one element in B which is not image of any element of A .

Example:

$$A = \{1, 2, 3\} \quad \& \quad B = \{a, b, c, d\}$$

$$f = \{(1, a), (1, b), (1, c)\}$$



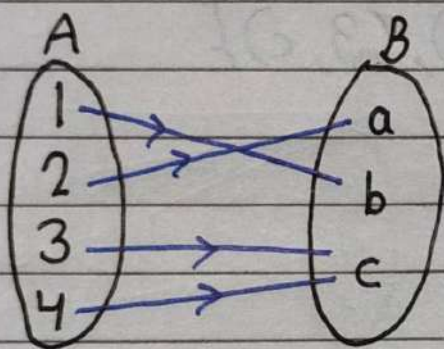
Onto (Subjective)

A function $f: A \rightarrow B$ is called onto function if every element of B has at least one pre-image.

Example:

$$A = \{1, 2, 3, 4\} \quad \& \quad B = \{a, b, c\}$$

$$f = \{(1, b), (2, a), (3, c), (4, c)\}$$



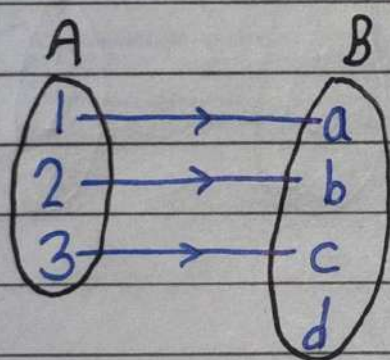
One-to-one Into Function

A function $f: A \rightarrow B$ is called one-to-one into function if distinct element of A are mapped with distinct element of B and also there exist at least one element in B which is not image of any element of A .

Example:

$$A = \{1, 2, 3\} \quad \& \quad B = \{a, b, c, d\}$$

$$f = \{(1, a), (2, b), (3, c)\}$$



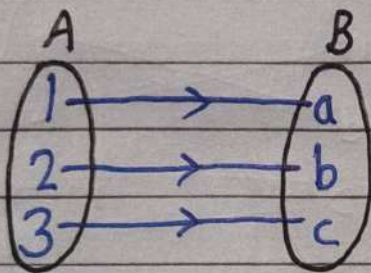
One-to-one Onto Element Function

A function $f: A \rightarrow B$ is called one-to-one onto function if it is one-to-one function and if every element of B has at least one pre-image.

Example:

$$A = \{1, 2, 3\} \quad \& \quad B = \{a, b, c\}$$

$$f = \{(1, a), (2, b), (3, c)\}$$



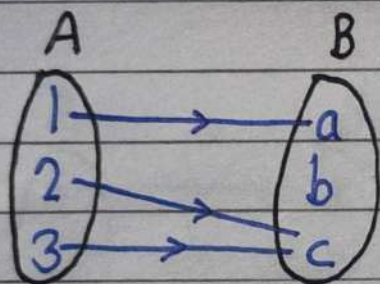
Many-to-one Into Function

A function $f: A \rightarrow B$ is called many-to-one into function if there exist two or more than two different element in A having same image in B , and if there exist at least one element in B which is not image of any element of A .

Example:-

$$A = \{1, 2, 3\} \quad \& \quad B = \{a, b, c\}$$

$$f = \{(1, a), (2, c), (3, c)\}$$



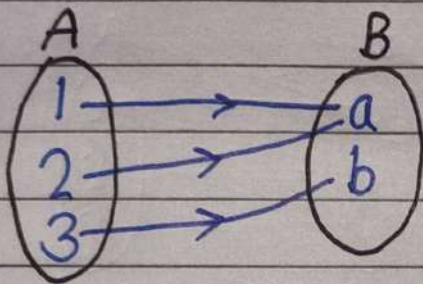
Many-to-one Onto Function

A function $f: A \rightarrow B$ is called many-to-one onto function if it is many-to-one function and if every element of B has at least one pre-image.

Example:

$$A = \{1, 2, 3\} \quad \& \quad B = \{a, b\}$$

$$f = \{(1, a), (2, a), (3, b)\}$$



Q. A function $f(x) = x^3 + 2$, check the given function is one-to-one or onto function?

Solⁿ:

For one-to-one function,
 $f(x_1) = f(x_2)$ — (1)

$$\text{Let, } f(x_1) = x_1^3 + 2$$

$$\& f(x_2) = x_2^3 + 2$$

Now, from eqⁿ. (1),

$$x_1^3 + 2 = x_2^3 + 2$$

$$\Rightarrow x_1 = x_2$$

So, the $f(x)$ is one-to-one function.

For onto function,
 $y = x^3 + 2$ — (2)

$$\Rightarrow y - 2 = x^3$$

$$\Rightarrow x = \sqrt[3]{y-2}$$

Putting the value of x in eqⁿ. (2),

$$y = (y-2) + 2$$

$$\Rightarrow y = y \quad \text{i.e. onto function.}$$

Operation on Function

It includes different operation such as: addition, subtraction, ~~multificat~~ multiplication and division.

If $f(x)$ & $g(x)$ are real-valued function, then

i. $(f+g)x = f(x) + g(x)$

ii. $(f-g)x = f(x) - g(x)$

iii. $(f \cdot g)x = f(x) \times g(x)$

iv. $\left(\frac{f}{g}\right)x = \frac{f(x)}{g(x)}, g(x) \neq 0.$

Q. If $f(x) = 9x - 5$ and $g(x) = 4x + 1$, then perform function operations?

Solⁿ:

$$(f+g)x = f(x) + g(x)$$

$$= 9x - 5 + 4x + 1$$

$$= 13x - 4.$$

$$(f-g)x = f(x) - g(x) = 9x - 5 - 4x - 1 = 5x - 6.$$

$$(f \cdot g)x = f(x) \times g(x) = (9x - 5) \cdot (4x + 1) = 36x^2 + 31x - 5.$$

$$(f.g)x = 36x^2 + 9x - 20x - 5$$

$$= 36x^2 - 11x - 5.$$

$$\left(\frac{f}{g}\right)x = \frac{f(x)}{g(x)} = \frac{9x-5}{4x+1} \quad \underline{\text{Ans.}} \quad \left[x \neq -\frac{1}{4}\right]$$

$$\text{Let, } \frac{9x-5}{4x+1} = 0$$

$$\Rightarrow 9x - 5 = 0$$

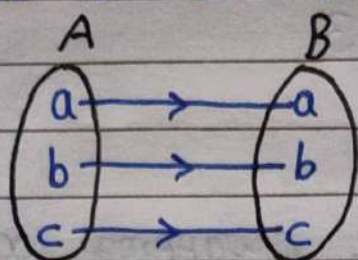
$$\Rightarrow x = \frac{5}{9}$$

If said find value.

Identity Function

$$f: A \rightarrow B$$

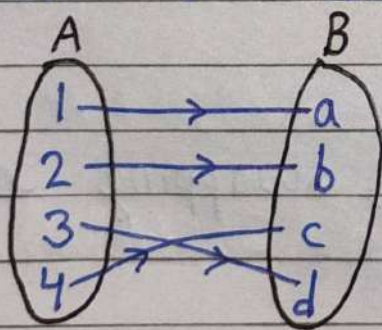
A function f is called identity function if each element of set A has a image on itself.



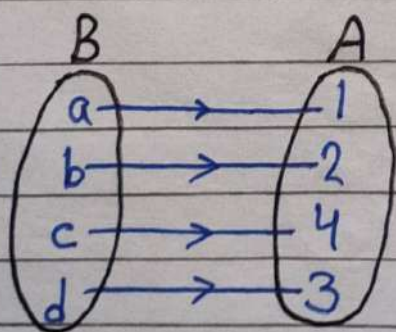
Inverse / Invertible Function

If function $f: A \rightarrow B$ is inverse function if it is bijective i.e. function is one-to-one; Thus, each element of A has ~~to~~ a distinct element of set B - and also function is onto; thus there is no element of B which is not the image of any element of A .

The inverse of function f is denoted by f^{-1} .



$$f: A \rightarrow B$$



$$f^{-1}: B \rightarrow A$$

Q. Find inverse of $f(x) = \frac{x-3}{2}$.

Solⁿ:

Let, $y = f(x) = \frac{x-3}{2}$

For inverse i.e. f^{-1} , swapping x & y ,

$$x = \frac{y-3}{2} \Rightarrow y = f^{-1}(x) = 2x + 3.$$

Q. Find inverse of $f(x) = \frac{3x+2}{4x-1}$, given $x=1$.

Solⁿ:

Given, $f(x) = \frac{3x+2}{4x-1} = y$ (let)

$$\Rightarrow f(1) = \frac{5}{3}$$

For inverse i.e. $f^{-1}(x)$, swapping x & y in $f(x)$,

$$x = \frac{3y+2}{4y-1}$$

$$\Rightarrow 4xy - x = 3y + 2$$

$$\Rightarrow y(4x-3) = x+2$$

$$\Rightarrow y = f^{-1}(x) = \frac{x+2}{4x-3}$$

$$f^{-1}(1) = 3/1 = 3.$$

Q. If $\mathbb{Q}$ is the set of rational number and $f: \mathbb{Q} \rightarrow \mathbb{Q}$, f is identity function and is a mapping define by $f(x) = 2x + 3$, $x \in \mathbb{Q}$. Proof that f is one-to-one and onto, also find f^{-1} ?

Solⁿ:

Given, $f(x) = 2x + 3$

For one-to-one function,
 $f(x_1) = f(x_2)$ — (1)

Let, $f(x_1) = 2x_1 + 3$
& $f(x_2) = 2x_2 + 3$

From (1), $2x_1 + 3 = 2x_2 + 3$
 $\Rightarrow x_1 = x_2$

So, $f(x)$ is one-to-one function.

For onto function,
let, $y = 2x + 3$ — (2)
 $\Rightarrow x = \frac{y-3}{2}$

Putting in (2),
 $y = 2\left(\frac{y-3}{2}\right) + 3 = y$

So, $f(x)$ is onto function also.

Also, $f^{-1}(x) = \frac{x-3}{2}$.

Composition / Product of Function

The composition of a function is an operation that takes two functions f & g and produces an function h , such that $h(x) = g(f(x)) = g \circ f$.

Example:

$$A = \{1, 2, 3, 4\}$$

$$B = \{a, b, c, d\}$$

$$f: A \rightarrow B = \{(1, b), (2, a), (3, c), (4, d)\}$$

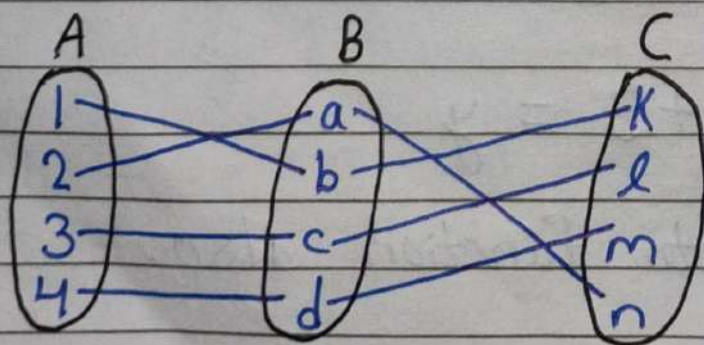
Also,

$$C = \{k, l, m, n\}$$

$$g: B \rightarrow C = \{(a, n), (b, k), (c, l), (d, m)\}$$

$$\therefore f \circ g = \{(1, k), (2, n), (3, l), (4, m)\}$$

$$\& g \circ f = \{(k, 1), (n, 2), (l, 3), (m, 4)\}$$



$$f: A \rightarrow B$$

$$g: B \rightarrow C$$

Q. If $f(x) = 2x + 1$ & $g(x) = x^2 + 3x$, find $f \circ g$ and $g \circ f$?

Solⁿ:

$$f \circ g = f(g(x)) = 2(x^2 + 3x) + 1 = 2x^2 + 6x + 1.$$

$$g \circ f = g(f(x)) = (2x + 1)^2 + 3(2x + 1)$$

$$\Rightarrow g \circ f = 4x^2 + 1 + 4x + 6x + 3$$

$$\Rightarrow g \circ f = 4x^2 + 10x + 4.$$

Q. If $f(x) = \frac{x}{x-1}$ & $g(x) = 1-x$, find $\frac{g \circ f}{f \circ g}$.

Solⁿ:

$$g \circ f = g(f(x)) = 1 - \frac{x}{x-1} = \frac{x-1-x}{x-1} = \frac{-1}{x-1}$$

$$f \circ g = f(g(x)) = \frac{1-x}{1-x-1} = -\left(\frac{1-x}{x}\right)$$

$$\frac{g \circ f}{f \circ g} = \frac{\left(\frac{-1}{x-1}\right)}{-\left(\frac{1-x}{x}\right)} = \frac{-1 \times -x}{(x-1)(1-x)} = \frac{x}{x-x^2-1+x}$$

$$\Rightarrow \frac{g \circ f}{f \circ g} = \frac{x}{-x^2 + 2x - 1}.$$

Q. Check the given function is inverse or not, if $f(x) = 2x+1$, $f: \mathbb{R} \rightarrow \mathbb{R}$ and $g(x) = x^2+3x$, $g: \mathbb{R} \rightarrow \mathbb{R}$.

Solⁿ:

Now,

$$f \circ g = f(g(x)) = 2(x^2+3x)+1 = 2x^2+6x+1$$

$$g \circ f = g(f(x)) = (2x+1)^2 + 3(2x+1)$$

$$\Rightarrow g \circ f = 4x^2+1+4x+6x+3$$

$$\Rightarrow g \circ f = 4x^2+10x+4$$

Now, putting $x=1$ in $f \circ g$,

$$f \circ g (x=1) = 2+6+1=9.$$

Now, putting $x=9$ in $g \circ f$,

$$g \circ f (x=9) = 4 \times 9^2 + 10 \times 9 + 4 = 418.$$

Hence, the given functions is not inverse.

Q. Let $f: \mathbb{N} \rightarrow \mathbb{N}$ defines $f(x) = 9x^2 + 6x - 5$, find f^{-1} and $f^{-1}(25)$. Also check the given function is inverse or not?

Solⁿ:

$$f(x) = 9x^2 + 6x - 5$$

$$\text{Let, } y = f(x) = 9x^2 + 6x - 5$$

Swapping x & y ,

$$x = 9y^2 + 6y - 5$$

$$\Rightarrow x + 5 = (3y + 1)^2 - 1$$

$$\Rightarrow x + 6 = (3y + 1)^2$$

$$\Rightarrow 3y + 1 = \sqrt{x + 6}$$

$$\Rightarrow y = f^{-1}(x) = \frac{\sqrt{x+6} - 1}{3}$$

$$\text{Now, } f^{-1}(25) = \frac{\sqrt{31} - 1}{3}.$$